

9.3 Operating Hours

The system will be operational for around 16 hours covering 2 shifts.

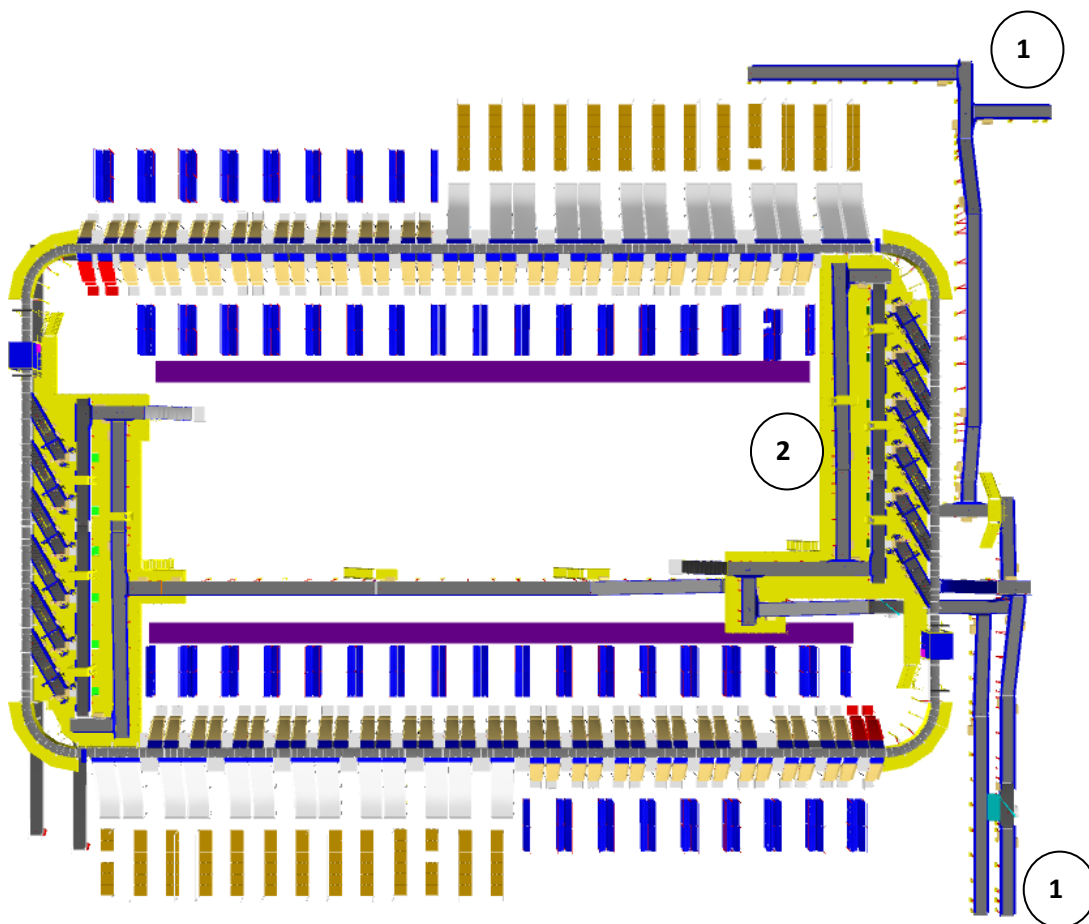
10. System description

10.1 Infeed System

The layout presents entire conveyors used as infeed conveyors for the shipments from market Place and Fulfilment Centre.

Each Zone relates to conveyor network to transport shipment volume to each Zones.

Contingency lines are also provided.



Legend

- 1 – Infeed PVC Conveyor for Marketplace and FC Load
- 2 – VDS Loop

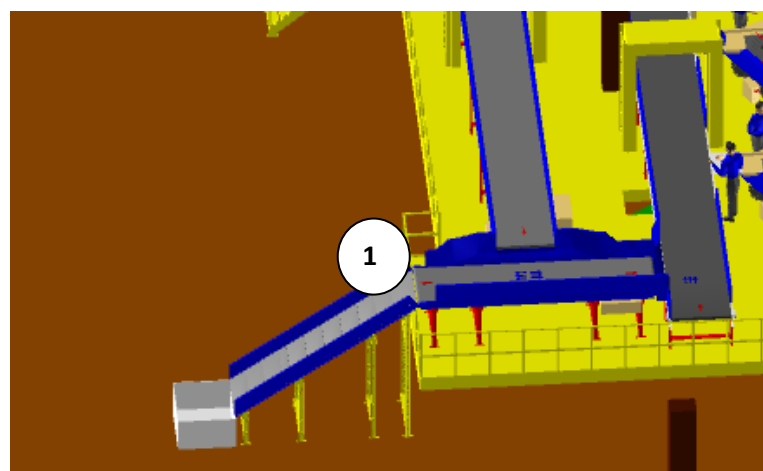
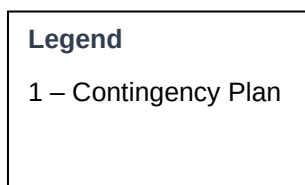
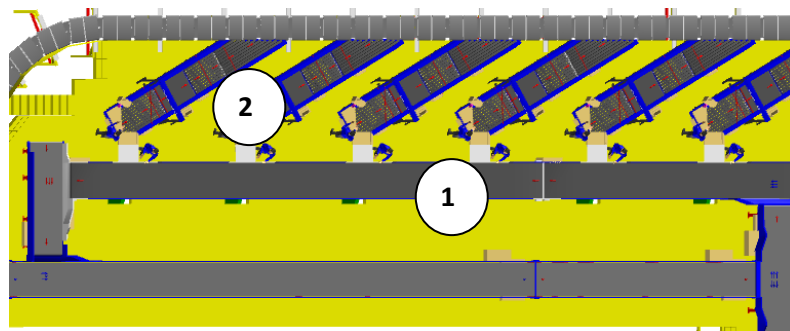
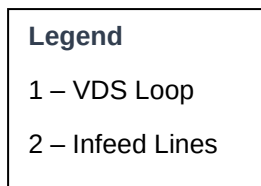
1. Infeed (Marketplace and FC)

- **Marketplace Load-** Representing 15% of the total load, these materials arrive from the docks (or marketplace). They are organized in temporary staging areas near the docks before being transferred via a conveyor system.
- **FC Load:** This constitutes the remaining 85% of the total load and dedicated conveyors transfers this load.

Bidirectional conveyors and Arm diverters are used for flexible load distribution between the two induct zones.

2. VDS Loop and Contingency Plan

VDS Loop is a conveyor-based arrangement in the form of a loop, where parcel move and is manually picked by an operator. It is developed to maintain the smooth operation by recirculating the parcel or product. The VDS loop restricts the parcels to accumulate and cause jams.



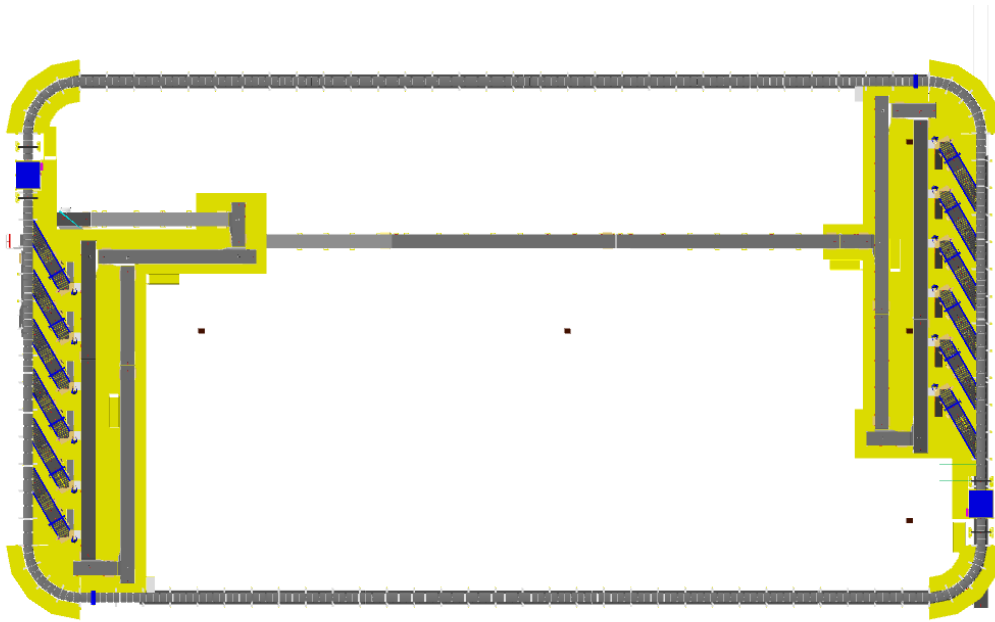
At VDS loops, Conveyors are planned which will run in reverse direction to transfer the load for manual sorting on Chutes.

3. Conveyor BOQ

S No.	Name	EL_1	EL_2	Conveyor Length (m)	Conveyor width (mm)	Set
1	Modular conveyor	3400	3400	10.0	1238	1
2	Modular conveyor	3400	3400	11.8	1238	1
3	Modular conveyor	3400	3400	17.6	1238	1
4	Modular conveyor	3400	3400	17.6	1238	1
5	PVC Belt Conveyor	1000	1000	19.7	1010	1
6	PVC Belt Conveyor	1000	1000	19.7	1010	1
7	PVC Belt Conveyor	888	3837	11.3	1210	1
8	PVC Belt Conveyor	800	800	19.7	1210	1
9	PVC Belt Conveyor	2400	2400	19.7	1210	1
10	PVC Belt Conveyor	550	550	7.2	1210	1
11	PVC Belt Conveyor	3544	3544	19.7	1210	1
12	PVC Belt Conveyor	800	800	7.2	1210	1
13	PVC Belt Conveyor	1050	1050	5.8	1210	1
14	PVC Belt Conveyor	2850	2850	3.8	1210	1
15	PVC Belt Conveyor	2521	3682	19.2	1210	1
16	PVC Belt Conveyor	2850	2850	3.8	1210	1
17	PVC Belt Conveyor	2521	3682	19.2	1210	1
18	PVC Belt Conveyor	3544	3544	19.7	1210	1
19	PVC Belt Conveyor	990	990	10.6	1210	1
20	PVC Belt Conveyor	803	1286	9.5	1210	1
21	PVC Belt Conveyor	2913	2913	3.0	1210	1
22	PVC Belt Conveyor	698	3164	9.7	1210	1
23	PVC Belt Conveyor	800	800	7.7	1210	1
24	PVC Belt Conveyor	1000	1000	8.2	1210	1
25	PVC Belt Conveyor	1000	1000	8.2	1210	1
26	PVC Belt Conveyor	3544	3544	2.9	1210	1
27	PVC Belt Conveyor	3600	3600	10.1	1210	1
28	PVC Belt Conveyor	3350	3350	4.8	1210	1
29	PVC Belt Conveyor	3600	3600	8.2	1210	1
30	PVC Belt Conveyor	3350	3350	13.0	1210	1
31	PVC Belt Conveyor	1300	1300	1.9	1210	1
32	PVC Belt Conveyor	687	1102	9.5	1210	1
33	PVC Belt Conveyor	3600	3600	4.3	1210	1
34	PVC Belt Conveyor	448	3625	13.0	1210	1
35	PVC Belt Conveyor	550	550	20.6	1210	1
36	PVC Belt Conveyor	2303	1410	5.0	1210	1
37	PVC Belt Conveyor	448	2480	7.8	1210	1
38	PVC Belt Conveyor	800	800	18.7	1210	1
39	PVC Belt Conveyor	1000	1000	5.3	1010	1
40	PVC Belt Conveyor	1000	1000	5.3	1010	1
41	Bagging Conveyor	300	400	19.9	1210	1

42	Bagging Conveyor	300	400	19.9	1210	1
43	Bagging Conveyor	300	400	19.9	1210	1
44	Bagging Conveyor	300	400	19.9	1210	1
45	Bagging Conveyor	238	338	19.9	1210	1
46	Bagging Conveyor	238	338	10.8	1210	1
47	Bagging Conveyor	300	400	13.7	1210	1
48	Bagging Conveyor	238	338	16.1	1210	1
49	Bagging Conveyor	238	338	19.9	1210	1
50	Bagging Conveyor	300	400	19.9	1210	1
51	Bagging Conveyor	300	400	15.6	1210	1
52	Bagging Conveyor	300	400	19.9	1210	1

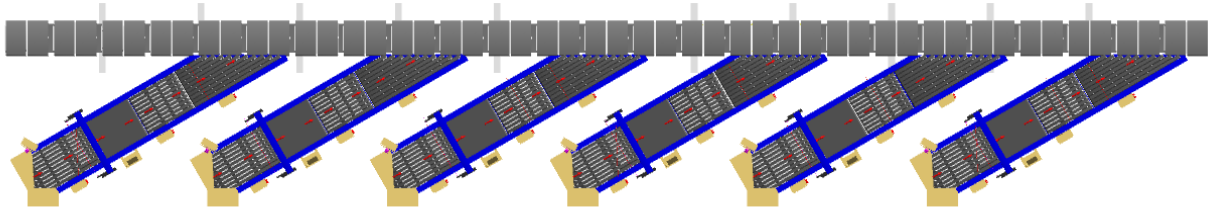
10.2 Induction to Sorter: Feedlines



The main purpose of the feedlines is to transport the shipments from the VDS Loop to the sorter by means Loading conveyors, Weighing, Buffer/Spacing Conveyor and Angle Merge conveyors. The flow of shipments on the Inducts are controlled by sensors in such a way that the required gap is ensured between the shipments.

These Feedlines are designed especially for Noon considering Operator Ease and ergonomics, where the operator ensures proper placement of the shipments on the Induct.

10.2.1 Feedlines



The Loop CBS has a total 12 Feedlines with Designed IPP of 2000 PPH.

1. Each Feedline has Four Conveyor Modules-
 - Orientation/Loading Conveyor - 1 Nos.
 - Weighing Conveyor- 1 Nos.
 - Buffer Conveyor- 1 Nos.
 - Intelligent Merge- 1 Nos.
2. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Shipment on the conveyor and prepare the Cross-Belt Cell for receiving the shipment with high precision.
3. Over the feedline, the shipments are evenly spaced and smoothly transferred to the main Loop.

1. Orientation/ Loading Conveyor –

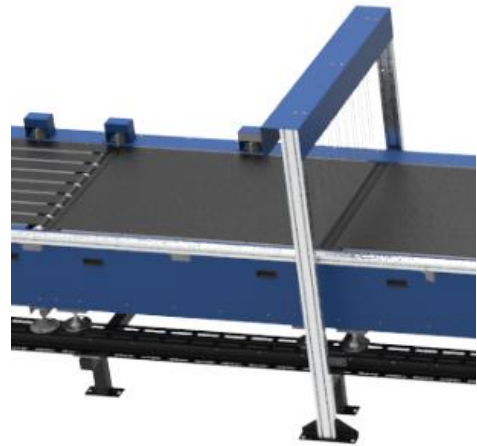
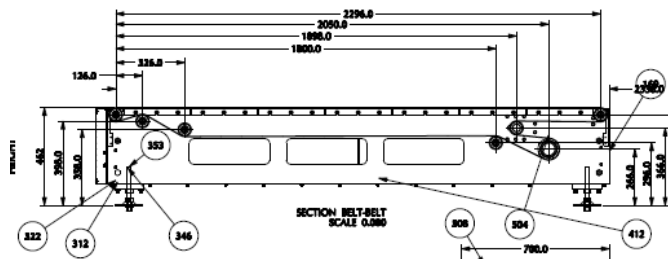
An Orientation conveyor is used to align and release the products for induction onto CBS. It serves as the initial point of entry where products are aligned and conveyed to subsequent conveyor modules. Associate aligns the parcels to the fixture mounted on loading conveyor and releases them. Irregular are dropped into nearby chutes to take them out of the system.



2. Buffer Conveyors

A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

This Conveyors required to maintain the Throughput of Line.

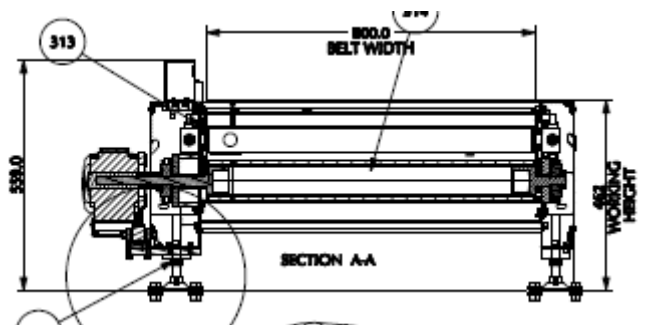


3. Weighing Conveyor

A Weighing Conveyor, also known as a Weigh Belt Conveyor, is a type of conveyor system specifically designed to measure the weight of materials as they move along the conveyor belt. It combines the functions of conveying and weighing into a single integrated process.

Weighing Conveyors equipped with high precision Load Cells to capture the weight of Shipments.

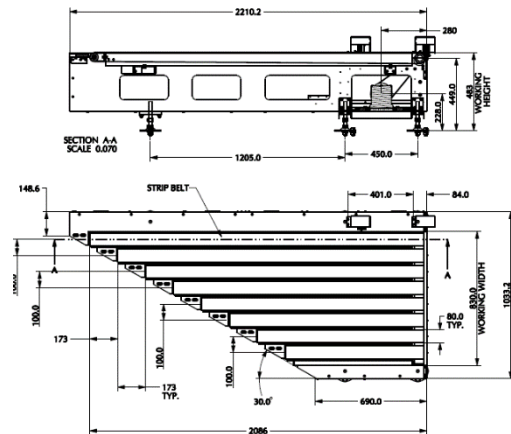
Make- Bizerba/ Mettler Toledo



4. Intelligent Merge Conveyor

An Intelligent Merge Conveyor incorporates advanced automation and control technologies to intelligently merge stream of materials into a single unified flow. It optimizes the merging process by dynamically adjusting the speed and position of items to ensure a smooth and efficient merge.

In the proposed solution this is 30° triangular high-speed conveyor used for inducting shipment/boxes directly on to the sorter. The Belts are Strip Belts for smooth shipment movement.

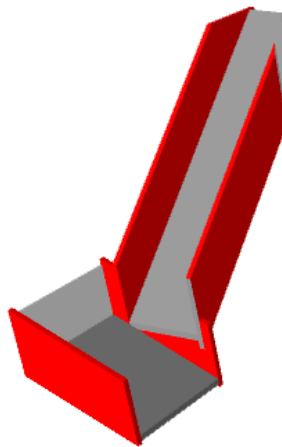


10.2.2 Irregular's Takeout Provision (Optional)

The system has a provision to take out irregulars by using combination of chutes and conveyors

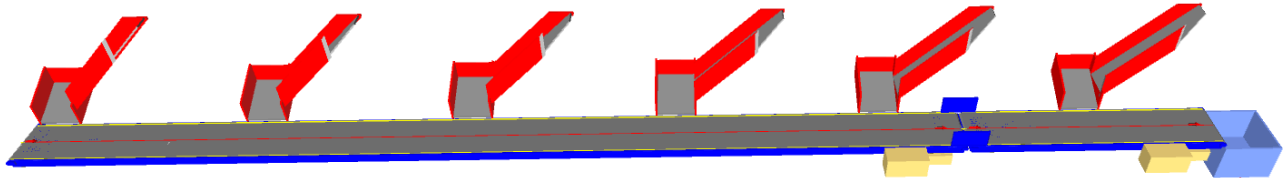
5. Irregular Chute (optional)

Specially designed chutes to ensure smooth flow of the shipments. Operator at inducts physically identify the irregular parcels and drop it into nearby irregular chute.



6. Irregular Takeaway Conveyor (optional)

This Conveyor network collect the shipments from all chutes and transport them to the collection bin.



7. Irregular Collection Bin (In Noon's scope)

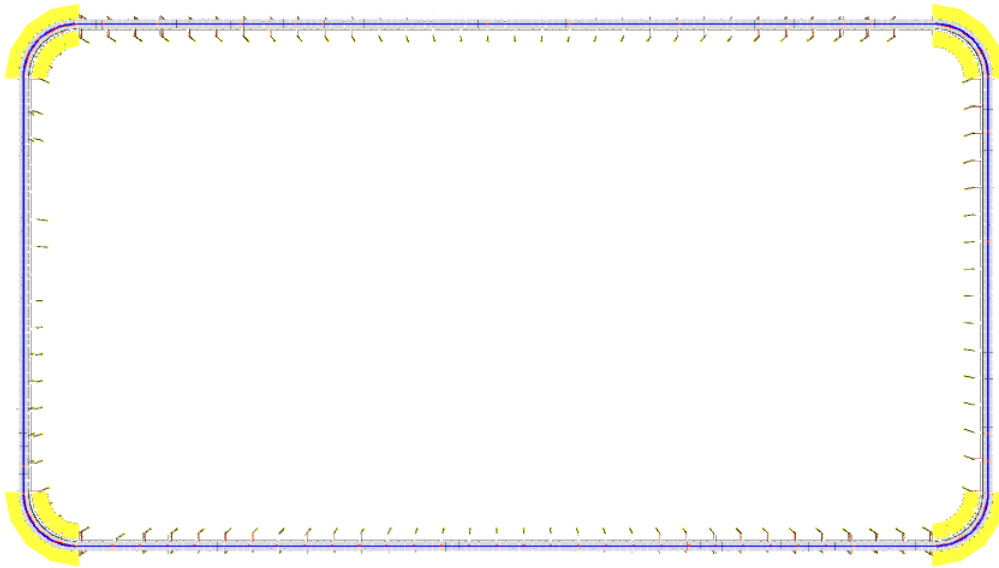
The shipments are collected in bins.



10.2.3 Conveyor BOQ (Optional-For Irregular Take away)

S No.	Name	EL_1	EL_2	Length (m)	width (mm)	Set
1	PVC Belt Conveyor	1000	1000	5.28	1010	1
2	PVC Belt Conveyor	1000	1000	5.28	1010	1
3	PVC Belt Conveyor	1000	1000	19.7	1010	1
4	PVC Belt Conveyor	1000	1000	19.7	1010	1

10.3 Main Loop

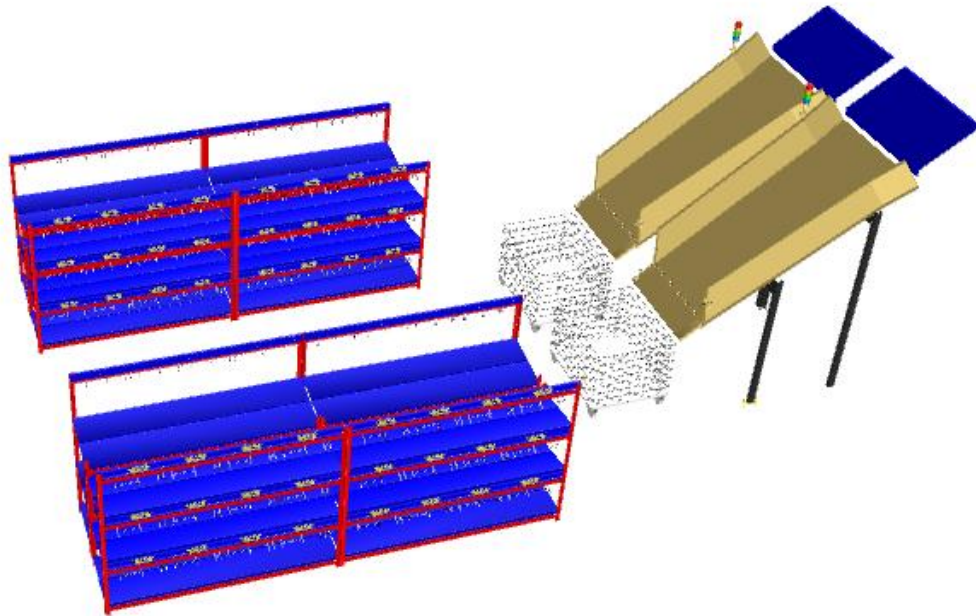


1. The proposed CBS Sorter to be installed will run at 3300 mm from ground level.
2. We have provided a loop with Dual Belt Carriers with 1175 mm pitch and Belt Size of 495 x 800mm.
3. One single loop CBS has a total length of 252 m & 215 carriers.
4. As Shipments pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the shipment reaches the respective chute, the carrier carrying the shipment for the chute will actuate and drop the shipment in the assigned chute.

10.4 Output Chutes

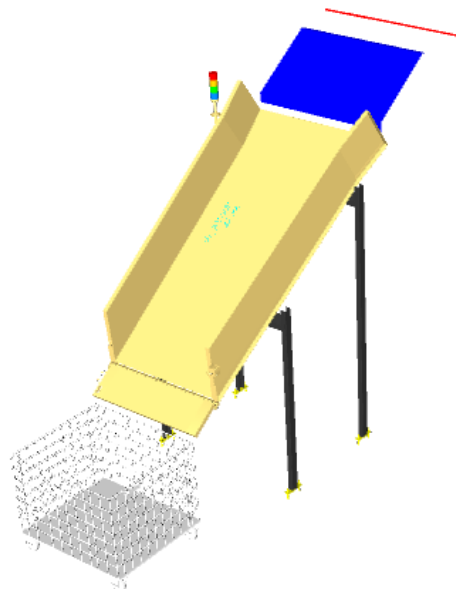
10.4.1 Sliding Chute + Secondary Sorting

A Sliding Chutes is used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes base metal to guide and support the flow of items along the chute & collect the shipments into trolleys placed at the end of each chute.



1. Sliding Chutes

Specially designed chutes ensuring smooth transfer of Parcels into collection bins.



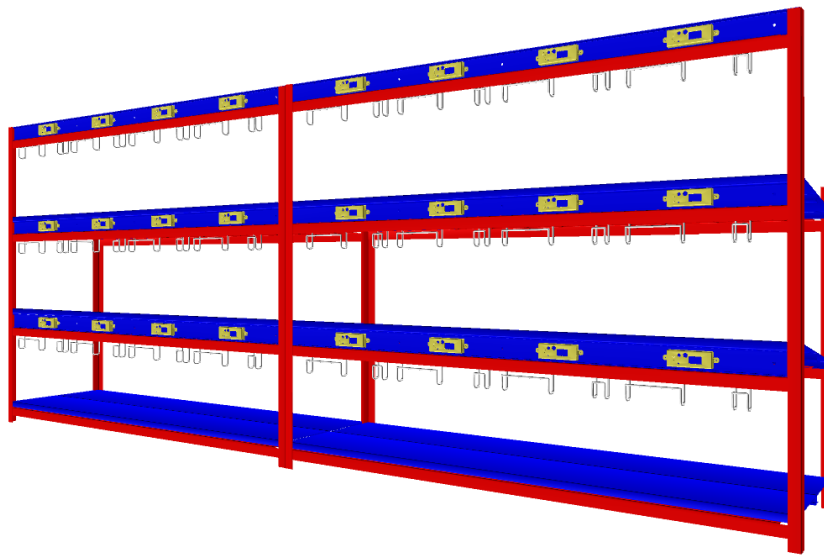
Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Four Color Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Four Colour Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations

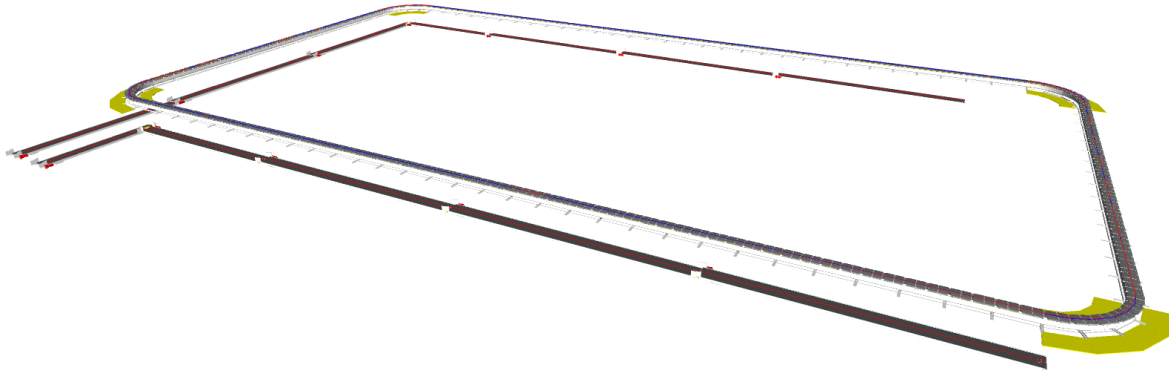
2. Bagging Type PTL

Shipments arriving through Sliding Chute accumulates into Roller Cage Trolleys. Then, they are manually scanned using a hand barcode scanner, then they are consolidated into Bags equipped in Bagging type PTL Racks.



3. Bag Take away Conveyor

In the Loop CBS, there are series of Bag Takeaway Conveyor to transport bags from sorter to a given location. Following the secondary sorting process, bags are securely fastened and manually placed onto these conveyors, which are positioned below the Loop CBS to cover all chute locations.



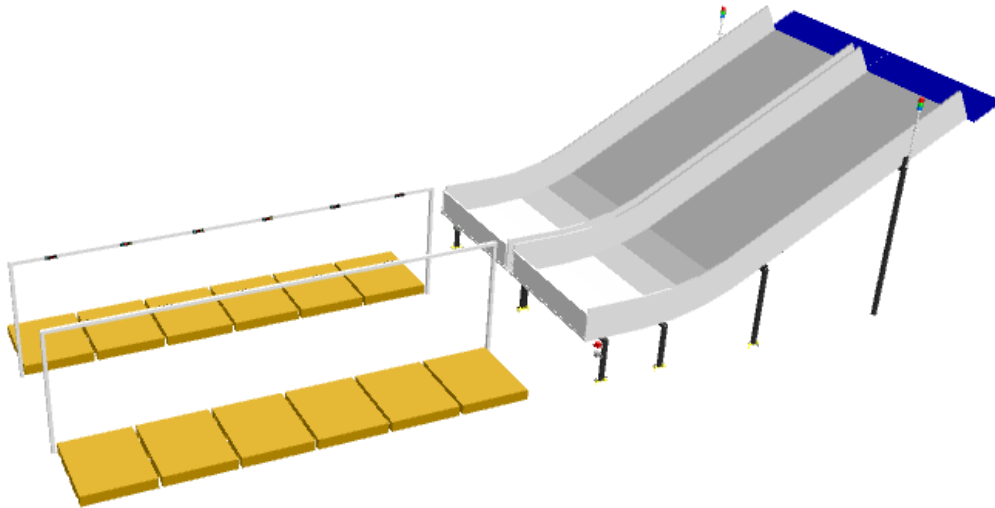
4. Conveyor BOQ (For Bag Take away)

S No.	Name	EL_1	EL_2	Length (m)	width (mm)	Set
1	PVC Belt Conveyor	300	400	19.9	1210	1
2	PVC Belt Conveyor	300	400	19.9	1210	1
3	PVC Belt Conveyor	300	400	19.9	1210	1
4	PVC Belt Conveyor	300	400	19.9	1210	1
5	PVC Belt Conveyor	238	338	19.9	1210	1
6	PVC Belt Conveyor	238	338	10.8	1210	1
7	PVC Belt Conveyor	300	400	13.7	1210	1
8	PVC Belt Conveyor	238	338	16.1	1210	1
9	PVC Belt Conveyor	238	338	19.9	1210	1
10	PVC Belt Conveyor	300	400	19.9	1210	1
11	PVC Belt Conveyor	300	400	15.6	1210	1
12	PVC Belt Conveyor	300	400	19.9	1210	1

10.4.1 Non-Sort Chutes + Secondary Sorting in Pallets

Non-Sort Chute is used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes base metal to guide and support the flow of items along the chute & have a collection portion to collect shipments in the chute.

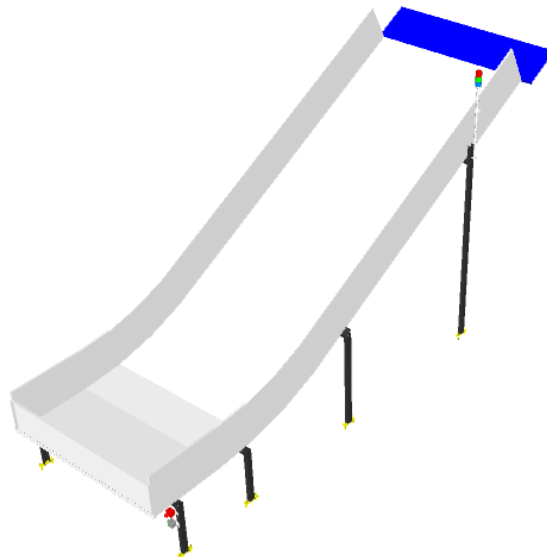
Secondary Sorting of parcels collected in chutes are done on pallets aided by PTL lights.



1. Non-sort Collection Chute

Non-Sort chute has capacity to accumulate 148 shipments, The sorted shipments will directly deposit into the collection portion of the chute

Calculation for Design: We have considered an average shipment size of (300 X 300 X 300 mm) while designing the capacity of chutes.



Average shipment size considered for chute volume calculation	L(m)	B(m)	H(m)	Volume (m3)
	0.3	0.3	0.3	0.027
Collection type PTL Chute	6.7	2	0.5	6.7

Assuming a 60% occupancy rate, about **148 shipments** will readily fit into these chutes.

Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Four Color Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Four Colour Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations

2. PTL set up for Palletizing

Shipments with Larger dimensions (that cannot be consolidated into bags), are palletized using PTL light system.



Image for representation purpose only

10.4.1 Rejection Chute

Rejection Chute is used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes base metal to guide and support the flow of items along the chute & collect the rejected shipments a Roller Cage Trolley.

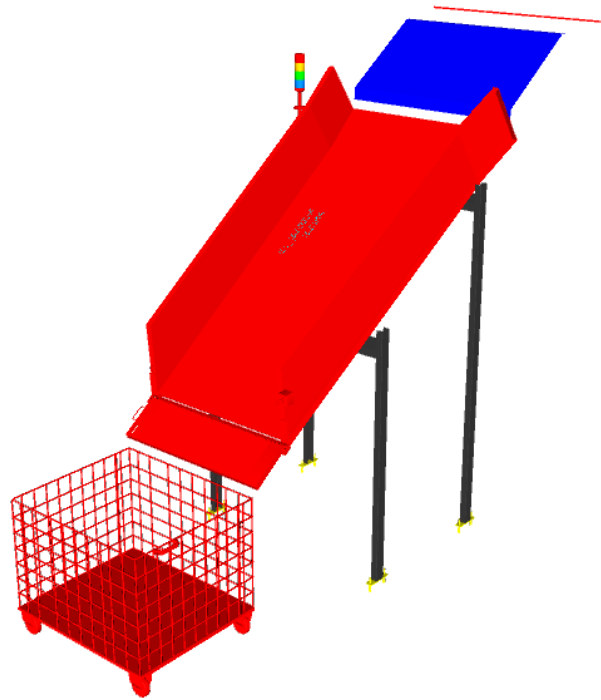


Image for representation purpose only

Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Four Color Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Four Colour Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations